MONASH University

FIT3181/5215 Deep Learning - S2 2026

KAGGLE COMPETITION REPORT

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Report

0. Set up
   Pytorch
   Pytorchvision

1. Dataset
   -Image resized to 72, random trimmed to 64x64 (noted, I check training data and some picture are smaller than 64)
   -20 classes
   -I didn't change the ratio left it at 85% train and 15% valid

2. Methods
   a. Model architecture

      Conv1 stem 3x3 stride 2: out 32x32
      Max pool 3x3 stride 2: out 16x16
      ResnextBlock: out 8x8

   b. Techniques (Maximum 150 words)
      - *Copy the set of data modified into cut-mix add to original set*
      - *Copy new set of data, random choice augment between rbg to gray scale, gaussian noise, and invert colour*

(I wanted more data because, I was trying to train a resnext model)

3. Experiment and results

Model:

- Tested modified resnext model:
  - -Dropped layer 4
  - -Change stem to 3x3
  - -Change number of resnext block with the following (original [3,4,6,3]: [2,2,2] -> [1,2,1] ->[1,1,1] -> [1,1]
  - - change cardinality from 32 to 16, to 8
- High 90 accuracy 0.3 loss in training, 70-90 accuracy in validation high loss ~1.0 , best 41 in testing.
- Test from 800000 parameter to 20,000 parameter.
- Test pre-trained ResNext50, retrained with 3x3 stem, fc layer 100 epoch -> High 90 and low loss in training, high 90 and high loss in validation, 41 in testing

b. Parameter Setting
(Hyper-parameters you tuned, your conclusions, and recommendations for the best settings to run your code.)

c. Loss & training accuracy
(Record your model performance on the training, validation and test set.)

4. Discussion and conclusion (Maximum 100 words)
Resnext model, even scaled down to 20,000 parameter by dropping layer 4, reduce layer stack to 1, change group/cardinality to 8 from 32 to 16. It was still hard to get valid loss to go down. I assume that the model is too complex and there is not enough data to reduce the loss. The model even when scaled down to 80,000 parameters which is similar to the model proposed in 3.1 was only able to achieved 41 in test slightly less than using pre-trained. However, the loss remained high for both model in validation. I was dubious on using pre-trained because it might contain the

The test filewas check and I saw some image was Gray scale, inverted color or blurry which is why I applied the transformation to the data. However, most of the data still look recognisable. I suspect that I needed to apply adversarial training. However, I was still unconfident in trying out adversarial training due to poor validation data on the training set.

5. References and Related works (If any)
(List articles/papers you have referred to or have used for your implementation.)
https://arxiv.org/abs/1611.05431

Submission Declarations

☐ I have provided accurate information in every detail of this report and in any attached documents.

❑ I understand that deductions for Assignment 1 will occur if I submit this report after the due date or provide any missing or incorrect information.
❑ I have submitted this report alongside my trained model. Evaluating my trained model on the test set yields the same (or almost exactly the same) accuracy as displayed on the Kaggle Leaderboard.
❑ I have compressed the files into a .zip extension and renamed it to *30750814_assignment01_report.zip*, where *<id>* is my student ID.